



OPEN Research on the impact of digital transformational leadership on digital innovation performance

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In the rapidly evolving digital economy, digital transformation has emerged as a crucial pathway for enterprises to achieve high-quality growth and maintain a competitive advantage. Enhancing digital innovation performance during the transformation process has become a pressing issue in both theoretical research and managerial practice. Grounded in Resource based View, this study examines the organizational transformation logic of “leader–strategy–performance” and develops a theoretical model that includes digital strategy as a mediating variable and digital transformation openness as a moderating variable. Based on data collected from 348 employees in China’s manufacturing sector, the study employed structural equation modeling to test the proposed hypotheses. The results reveal that (1) digital transformational leadership significantly enhances employees’ digital innovation performance; (2) digital strategy partially mediates this relationship; (3) digital transformation openness positively moderates the link between digital strategy and digital innovation performance; and (4) it also amplifies the moderated mediation effect of digital transformational leadership on digital innovation performance via digital strategy.

Keywords Digital transformational leadership, Digital strategy, Digital innovation performance, Digital transformation openness

Amid rapid advancements in digital technologies, the rise of big data, cloud computing, and the Internet of Things is driving human society into a new digital era¹. Digital transformation has become a strategic priority for achieving high-quality development, and how to advance it effectively has become a major concern in the business world². It is anticipated that globally 85% of organizations will be digitalized and 42% will be fully automated by 2027³. Digital transformation is now a vital pathway for the survival and long-term development of traditional enterprises. Digital innovation performance is increasingly seen as a key indicator for assessing the success of digital transformation⁴. Employees as the direct participants and executors of digital innovation, play a crucial role in shaping a firm’s overall digital innovation performance through their behavior and outcomes in digital contexts⁵. Although employee-level digital innovation performance is widely acknowledged as important, the specific mechanisms shaping employees’ capacity to produce innovative outcomes in digital contexts remain insufficiently understood. This gap highlights the need to explore how organizational factors can be leveraged to enhance employee digital innovation performance and facilitate successful digital transformation.

The Resource Based View (RBV) is a widely applied theoretical perspective in strategic management. It is grounded in an “inside-out” analytical framework, highlighting that a firm’s long-term success originates from its distinctive internal resources and capabilities. These resources encompass not only tangible assets but, more critically, intangible ones such as knowledge, organizational culture, and leadership, which are difficult to replicate⁶. According to the RBV, firms gain competitive advantage by possessing resources and capabilities that are unique, valuable, rare, and difficult to imitate^{6,7}. In the context of digital transformation, human resources remain central to managerial decision-making and strategic transformation in all organizations⁸. As individuals with distinct roles and authority, leaders play a pivotal role in guiding organizational transformation and development. In the digital era, transformational leaders must integrate digital technologies and embody digital attributes to meet the evolving demands of leadership roles and effectively lead digital transformation. As a result, the concept of digital transformation leadership has emerged⁹. Digital transformation leadership constitutes a unique managerial resource and dynamic capability that is essential for achieving competitive advantage in the digital era¹⁰.

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The effective deployment of leadership capabilities in digital transformation depends on strategic coordination and resource orchestration. As a core element of complex enterprise management systems, digital strategy serves as a comprehensive blueprint for value creation and has long been a central concern in management research^{11,12}. From the RBV, digital strategy is not only a guiding framework for resource allocation but also a critical mechanism through which enterprises convert diverse digital resources into unique capabilities¹³. Digital strategy acts as a bridge between leadership vision and operational execution, enabling organizations to systematically integrate digital technologies into their business models, processes, and value propositions. The key criterion for evaluating digital innovation performance is whether an enterprise can sustain or even enhance its value creation capacity following transformation¹⁴. Enterprises undergoing digital transformation must develop new digital strategies aligned with the demands of the digital age to fully harness the benefits of transformation and enhance the tangible outcomes of digital innovation. However, existing research has not adequately examined how digital transformational leadership translates into improved employee digital innovation performance through strategic pathways.

During the implementation of digital strategies, organizational members often display diverse attitudes toward change. The success of digital transformation depends not only on technology deployment and leadership behaviors but also on employees' psychological readiness and cognitive acceptance¹⁵. According to the RBV, employee's openness is a vital human resource for the enterprise, and the quality of this resource directly influences the effectiveness of digital transformation strategy implementation. This openness reflects the extent to which organizational members support digital integration, demonstrated by proactive tool adoption, active participation in technological learning, and adaptation to new work methods¹⁶. Higher levels of openness reduce employee resistance, thus improving the efficiency of digital strategy implementation and strengthening the organization's ability to convert efforts into performance outcomes¹⁷. Therefore, incorporating digital transformation openness into the analytical framework not only emphasizes the role of human factors in strategy implementation but also uncovers deeper interaction mechanisms within the leadership, strategy, and performance pathway.

Building on the foregoing discussion, this study seeks to address two central research questions. First, how does digital transformational leadership influence employee digital innovation performance through strategic mechanisms? Second, under what conditions is this strategic pathway most effective in driving employee digital innovation performance? To address these questions, grounded in the RBV, this study conceptualizes digital transformational leadership as a critical managerial resource that enhances employee digital innovation performance by enabling the formulation and effective implementation of digital strategy. Furthermore, by incorporating digital transformation openness as a key boundary condition, this study emphasizes the importance of employees' proactive engagement with and adaptive capacity for digital transformation. By integrating these elements, this study elucidates the specific strategic pathways and boundary conditions through which digital transformational leadership drives employee digital innovation performance.

Theory and hypotheses

Resource based view (RBV)

The Resource-Based View (RBV) represents a fundamental theoretical framework in strategic management that emphasizes how organizations achieve competitive advantage through the strategic deployment of valuable, rare, inimitable, and non-substitutable (VRIN) resources and capabilities, and goes beyond mere possession to highlight resource orchestration through the structuring, bundling, and leveraging of resources to create value^{6,7}. Within this framework, resources encompass not only tangible assets such as financial capital and physical infrastructure, but also intangible assets including knowledge, capabilities, organizational culture, and leadership competencies^{18,19}. The dynamic interplay between resource accumulation, capability development, and strategic deployment determines an organization's ability to create and sustain competitive advantages in rapidly changing environments²⁰.

This study develops a theoretical model based on the RBV, emphasizing the pathway from digital transformation leadership to strategy and performance. Within this framework, digital transformation leadership is viewed as a distinctive and valuable managerial resource. As core human capital, it enables firms to integrate and apply other essential resources efficiently. Digital transformational leaders act as resource orchestrators by identifying valuable digital resources and guiding organizational members in strategic initiatives, fulfilling the VRIN criteria through their distinctive combination of digital expertise and transformational competencies. Accordingly, digital strategy serves as the primary mechanism through which leadership transforms its capabilities into tangible outcomes. It represents the bundling process in which diverse digital resources are systematically integrated into coherent organizational capabilities. Digital strategy involves structured planning that enables firms to leverage these resources effectively to redesign business models and drive innovation. Digital transformation openness is also a human resource. This proactive mindset and openness to change not only enhance the efficiency of strategy execution but also constitute a vital aspect of the organizational climate that determines whether resources can be effectively transformed into innovation performance outcomes.

Digital innovation performance

Digital innovation performance has emerged as a central topic in recent research on corporate management and innovation, with scholars systematically exploring its definition and evaluation criteria. Drucker²¹ was among the first to highlight that innovation performance, in the context of R&D and innovation activities, reflects an enterprise's development status and future growth potential. Coombs²² further emphasized that innovation performance is a comprehensive, outcome-oriented indicator. It captures not only the inputs and outputs during the technology development phase but also the knowledge, capabilities, and experience accumulated in the process, making it a crucial measure of R&D effectiveness. Tierney and Farmer²³ defined innovation performance,

from a product-outcome perspective, as the creation of valuable products, services, or processes within a specific domain. Wang and Zhang²⁴ broadened the concept of innovation performance by integrating digital technologies, including not only product innovation but also business model innovation and organizational digital transformation. While these perspectives clarify innovation performance at the organizational level, they often overlook individual-level behaviors and contributions. This limitation is especially relevant in the context of digital transformation, where individual employee roles are increasingly vital in driving organizational innovation.

To overcome the limitations of previous research, this study redefines digital innovation performance from the viewpoint of individual employees, building on the work of Janssen and Zhou^{25,26}. Digital innovation performance is conceptualized as the extent to which employees, in digital work contexts, intentionally implement creative ideas, actively use digital tools to demonstrate creativity and problem-solving skills, enhance their competitiveness through concrete innovative behaviors, and ultimately generate visible outcomes and value. This definition underscores not only employee's initiative and creativity in the innovation process, but also the pivotal role of digital technologies as enablers, and the strong connection between individual and organizational innovation outcomes. Thus, digital innovation performance extends beyond traditional dimensions such as product or technological outputs. It emphasizes outcomes generated by employees through knowledge integration, problem-solving, and value creation in digital environments. This perspective offers substantial theoretical and practical value for understanding employee behavior amid digital transformation²⁷.

Digital transformational leadership and digital innovation performance

As technology evolves and the external business environment continues to change, the digital adaptability of organizational leaders has become an increasingly critical driver of successful transformation³. From the perspective of individual leadership traits, digital transformational leaders equipped with digital thinking can accurately anticipate technological trends, define clear strategic directions and investment priorities, and guide their organizations toward enhanced digital performance²⁸. These leaders inherit the strengths of traditional transformational leadership, such as motivation, vision-setting, and organizational empowerment while also integrating core elements of digital leadership, including technological sensitivity, data-driven decision-making, and agility in innovation²⁹.

When making strategic decisions, digital transformational leaders typically exhibit insightful and forward-looking judgment. This enables them to identify and seize opportunities arising from digital advances, leading their firms to either transform existing business models or establish entirely new digital ecosystems³⁰. Digital transformational leadership influences digital innovation performance through multiple interrelated mechanisms. First, such leaders foster an innovation-friendly organizational climate by encouraging employees to experiment, take risks, and engage in continuous learning. They create a psychologically safe environment that empowers employees to explore novel digital solutions and challenge existing paradigms without fear of failure or punishment. Second, digital transformational leaders excel in mobilizing and allocating resources, ensuring that innovation initiatives have the necessary technological infrastructure, financial support, and human capital required for successful implementation, thereby enhancing employee innovation performance³¹. Therefore, we propose the following hypothesis:

Hypothesis 1 Digital transformational leadership is positively related to digital innovation performance.

Digital strategy and digital innovation performance

A digital strategy refers to a strategic initiative through which firms respond to rapidly evolving digital environments by leveraging advanced technologies such as big data, cloud computing, social media, and the Internet of Things to systematically redesign their business models, product architectures, and organizational operations¹². Digital strategy encompasses not only the adoption of technological tools but, more fundamentally, the enterprise's capability to achieve process automation and organizational restructuring through digital technologies³². At its core, digital strategy is a systematic process through which organizations integrate and reconfigure digital resources across key domains such as business models, value creation logic, process design, and organizational management to drive transformation and sustained innovation¹³. As digital technologies become increasingly prevalent, the digital transformation of traditional manufacturing firms has shifted from a purely technical concern to a central issue in strategic management. Digital strategy plays a pivotal role in enhancing firms' digital innovation performance. First, the systematic implementation of digital strategy enhances firms' capabilities to leverage technology for product innovation, process optimization, and service redesign, thereby improving overall innovation outcomes³³. Second, by restructuring organizational frameworks, operational processes, and resource allocations, firms can develop more agile and intelligent operational systems, thereby strengthening their internal R&D capabilities and responsiveness to market dynamics.

Moreover, a digital strategy can enhance information connectivity and promote collaborative innovation among firms and key stakeholders throughout the value chain. Digital strategy promotes collaboration between firms and external partners including suppliers, customers and research institutions³⁴. This collaboration facilitates resource integration across organizational boundaries and drives joint innovation, thereby improving digital innovation performance in products and services³⁵. This form of digital collaboration helps technology-driven firms achieve high-quality innovation outcomes in product development, service design and process optimization. Therefore, we propose the following hypothesis:

Hypothesis 2 Digital strategy is positively related to digital innovation performance.

Mediating effect of digital strategy between digital transformational leadership and digital innovation performance

Digital innovation performance is often stimulated by technological change. However, technological advancement alone is not sufficient to ensure sustained improvements in a firm's digital innovation outcomes³⁶. During digital transformation, firms need digital transformational leaders who possess strategic vision and the capacity to drive change. These leaders must be able to coordinate comprehensive transformation across internal structures, organizational culture and the external business environment³². In this context, digital strategy plays a central role in a firm's digital innovation efforts. It functions not only as a structured framework for implementing digital technologies but also as a vital link that transforms innovation achievements into competitive advantage¹³.

Digital transformational leadership plays a pivotal role in shaping the development and implementation of a firm's digital strategy. It drives cultural change, supports process redesign, and facilitates business model restructuring, thereby improving the effectiveness of digital strategy execution¹⁷. Acting as a key mediating mechanism between leadership and innovation performance, digital strategic initiatives deepen digital innovation implementation and elevate the firm's overall level of digital transformation³³. This process facilitates the integration of digital technologies into core business operations and enables the effective conversion of digital innovation outcomes into sustainable innovation returns.

In strategic practice, digital transformational leaders adopt a disruptive innovation mindset. They focus on the feasibility of strategy execution and the redesign of business logic to identify and capture new opportunities for organizational transformation³⁷. By building an innovative and adaptive digital strategy system, firms can drive fundamental changes in their organizational structures, operational models, and value creation mechanisms. This enables entry into new markets, expansion of customer bases, and diversification of value co-creation channels^{38,39}. Furthermore, by actively building and managing digital platforms and ecosystems, firms can use digital technologies to facilitate more efficient and higher-quality value interactions. This allows them to extensively share and absorb external benefits generated through digital innovation^{40,41}. This ecosystem-oriented strategic mindset fundamentally strengthens a firm's digital innovation performance. Therefore, we propose the following hypothesis:

Hypothesis 3 Digital strategy mediates the relationship between digital transformational leadership and digital innovation performance.

Moderating role of digital transformation openness

Digital transformation is not merely a matter of technology adoption but a comprehensive organizational change involving the restructuring of management processes, innovation of business models, adjustment of organizational structures, and reconstruction of corporate culture⁴². Therefore, the successful implementation of digital strategies depends not only on the establishment of technological infrastructure but also on organizational members' understanding and acceptance of the impacts of digital technological change⁴³. In this context, digital transformation openness is regarded as a key moderating factor influencing the effectiveness of digital strategy implementation. This concept refers to employees' positive attitudes, cognitive understanding, and behavioral intentions toward digital transformation, reflecting their psychological and behavioral acceptance and support for the integration of digital technologies⁴⁴. Specifically, this openness manifests in employees' willingness to adopt new digital tools, actively participate in technological change, and continuously learn and adapt to the evolving digital environment¹⁶.

According to the RBV, digital transformation openness is a vital organizational resource. It shapes a firm's capacity to identify, assimilate, and apply digital knowledge. Such openness functions as a complementary asset. It increases the effectiveness of digital strategy implementation by reducing internal resistance and facilitating resource integration. Organizations with high levels of digital transformation openness tend to exhibit stronger dynamic capabilities. These capabilities allow for flexible adjustments to daily operations and processes in response to digital strategic initiatives⁴⁵. The moderating role of digital transformation openness aligns with a central RBV principle: the value of strategic resources is contingent on the existence of complementary assets and a supportive organizational context⁷. When employees demonstrate greater openness to digital transformation, digital strategies are executed more effectively. This openness fosters internal conditions that support resource reconfiguration and capability development⁴⁶, enabling employees to better leverage organizational resources to produce innovative outcomes. In other words, digital transformation openness moderates the relationship between digital strategy and digital innovation performance: under conditions of high openness, digital strategies are more likely to generate tangible and valuable innovative outcomes. Therefore, we propose the following hypothesis:

Hypothesis 4 Digital transformation openness positively moderates the relationship between digital strategy and digital innovation performance, such that the relationship becomes stronger when digital transformation openness is high rather than low.

Moderated mediating effect

Hypothesis 3 and 4 together form a mediation model with moderating, which is based on the moderating mediator inference method⁴⁷. Digital transformation openness moderates the relationship between digital strategy and digital innovation performance. This study further predicted that digital transformation openness positively moderates the mediating effects of digital transformational leadership on digital innovation performance via digital strategy. Therefore, the following hypothesis is proposed.

Hypothesis 5 Digital transformation openness positively moderates the indirect effect of digital transformational leadership on digital innovation performance through digital strategy. In other words, the higher the digital transformation openness, the greater the mediating effect of digital strategy.

Figure 1 shows the theoretical model developed in this study.

Materials and methods

Sample and procedures

Data for this study were collected from March 2025 to April 2025. The research sample mainly included employees from manufacturing firms that are implementing digital transformation in the Shandong province of China. Shandong Province was chosen as the research sample because it is one of China's leading manufacturing hubs and a pilot region for national digital transformation initiatives. The province hosts numerous traditional manufacturing firms that actively adopt digital technologies, making it a highly representative region for examining digital transformation in China's manufacturing sector. To ensure sample authenticity, the following criteria were applied: (1) the firm has formally initiated a digital transformation plan; (2) the firm has invested in digital technologies such as big data, cloud computing, robotics, or the industrial internet; (3) the firm has implemented at least one organizational or process change related to digitalization, such as a digital supply chain or smart manufacturing). The validity of these criteria was verified using official corporate reports, website information, and confirmation from HR managers during preliminary communications. Among the 25 companies approached a total of 7 entities accorded their consent to participate in the study. A formal survey was conducted using a web-based questionnaire. Approval was guaranteed by the relevant human resource heads of the companies, who willingly participated in the surveys. This study separated the independent variables from the dependent variables in survey waves to mitigate common method bias⁴⁸. The questionnaire survey comprised two stages: During Time 1 (T1), employees completed questionnaires regarding a predictor variable (digital transformational leadership), a mediating variable (digital strategy), and demographic variables (age, gender, education, seniority and Years implementing DT). After a month, during Time 2 (T2), the same participants completed questionnaires regarding a moderating variable (digital transformation openness) and a dependent variable (digital innovation performance). To match the responses obtained during T1 and T2, participants were asked to enter the last four digits of their ID numbers in the questionnaires.

A total of 560 questionnaires were distributed in this study and 407 questionnaires were collected. After filtering out incomplete responses and removing outliers, 348 valid questionnaires remained with a response rate of 62.1%. In terms of the sample distribution, 175 (50.29%) respondents were males and 173 (49.71%) were females. Most of the respondents were aged between 26 and 30 years (39.08% of the total survey population). The majority possessed a bachelor's degree (49.43% of the total survey population), seniority ranged mostly between 4 and 6 years (28.44% of the total survey population) and years implementing DT ranged mostly between 1 and 2 years (27.30% of the total survey population) (Table 1). The sample size was adequate to analyze the model. Using the rule of thumb of (no. of items in questionnaire $\times 5 = 24 \times 5$) which is 120⁴⁹. Therefore, the sample of 348 is adequate because this is more than the required number of 120 responses. To further assess the adequacy of our sample size beyond the item-ratio criterion, we conducted a post hoc power analysis using G*Power 3.1. With a significance level of 0.05, a medium effect size ($f^2 = 0.15$), and nine predictors, the analysis yielded a statistical power of 0.99. This substantially exceeds the commonly recommended threshold of 0.80, indicating that our sample of 348 respondents provides sufficient power to detect the hypothesized mediation and moderation effects.

Measures

The scales used in this study were mainly derived from mature scales used in the academic community, with proven reliability and validity in domestic and foreign studies. All scales used the 5-point rating like the Likert

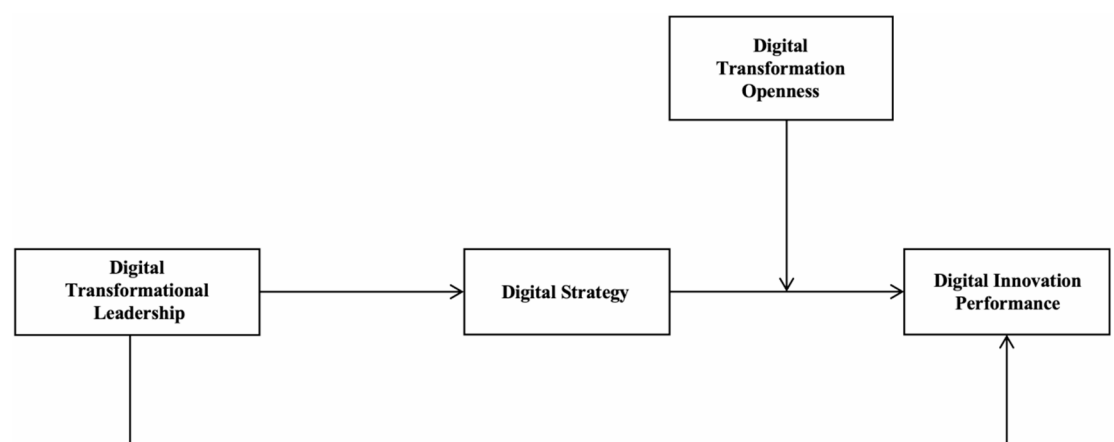


Fig. 1. Theoretical model.

Variable		N	Percentage			N	Percentage
Gender	Male	175	50.29	Seniority	≤ 1	65	18.68
	Female	173	49.71		[1, 3]	91	26.15
Age	[20, 25]	49	14.08		[4, 6]	99	28.44
	[26, 30]	136	39.08		[7, 10]	81	23.28
	[31, 35]	94	27.01		≥ 10	12	3.45
	[36, 40]	44	12.64		≤ 1	50	14.37
Education	≥ 40	25	7.18	Years implementing DT	[1, 2]	95	27.30
	High school	10	2.87		[3, 4]	86	24.71
	Associate degree	73	20.98		[5, 6]	68	19.54
	Bachelor degree	172	49.43		≥ 6	49	14.08
	Master degree	68	19.54				
	doctoral degree	25	7.18				

Table 1. Demographics of the survey respondents. (N = 348).

scale, where 1 means strongly disagree and 5 means strongly agree. The specific measurement of each variable is as follows.

Digital transformational leadership

Digital transformational leadership was measured with the 6-item scale developed by Alnuaimi et al.⁹. The items are as follows: “Our leaders inspire all members with the digital transformation plans for our organization,” “Our leaders provide a clear digital transformation vision for the organization’s members to follow” etc. Cronbach’s alpha for this scale was 0.874.

Digital strategy

Digital strategy was measured with the 4-item scale developed by Li et al.⁵⁰. The items are as follows: “In my organization, we integrate digital technology and business strategy to attain strategic alignment with the government and other partners,” “In my organization, we create a shared vision of the role of digital technology in business strategy” etc. Cronbach’s alpha for this scale was 0.867.

Digital innovation performance

8-item were adapted and modified to fit the digital transformation context from previous employee innovation performance scales (Chang et al.⁵¹ to measure digital innovation performance. The items are as follows: “I manage to implement my innovative ideas involving digital technologies at work,” “I transform innovative ideas into useful digital applications or tools.” etc. Cronbach’s alpha for this scale was 0.916.

Digital transformation openness

For the measurement of digital transformation openness, 6-item were adapted from Hinduan et al.⁵². The items are as follows: “I am very interested in using digital technologies and digital tools,” “I show strong willingness to learn new knowledge and digital technologies” etc. Cronbach’s alpha for this scale was 0.892.

Results

All statistical analyses were conducted using IBM SPSS Statistics (Version 29.0; <https://www.ibm.com/products/spss-statistics>), Mplus (Version 8.0; <https://www.statmodel.com>), and SmartPLS (Version 4.1; <https://www.smartpls.com>). Specifically, SEM was run with Mplus and SmartPLS to test reliability, validity, and model fit. To examine the hypotheses, this study performed partial least squares structural equation modelling (PLS-SEM) using SmartPLS.

Reliability and validity

We assessed the constructs’ reliability using the internal consistency measure analysis, obtaining acceptable values for Cronbach’s alpha and composite reliability (CR). However, the factor loadings of items DTL6 and DTO4 in the DTL and DTO constructs were both below 0.5, falling short of the acceptable threshold⁵³. Therefore, we dropped item DTL6 and DTO4 to improve the model’s internal consistency and reliability and avoid having an issue with the model. A post hoc inspection suggests that the poor performance of these items may have resulted from translation nuances or cultural interpretations, rather than from fundamental flaws in the underlying constructs. Table 2 summarizes the results of convergent validity and internal consistency reliability. All indicators and constructs meet the required measurement criteria. Specifically, the factor loadings are all above 0.759, demonstrating that indicator reliability is achieved⁵⁴. In addition, Cronbach’s alpha value of each construct ranged from 0.867 to 0.916 (exceeding 0.7). The AVE values ranged from 0.630 to 0.715 (exceeding 0.5), denoting that convergent validity is also achieved⁵⁵. Furthermore, CR values are 0.908 to 0.932, well above the required minimum level of 0.70, thus demonstrating internal consistency⁵⁶. In other words, the results show that the model has good convergent validity and internal consistency.

Items	Loading	Ca	CR	AVE
DTL		0.874	0.908	0.664
Item1	0.779			
Item2	0.835			
Item3	0.840			
Item4	0.813			
Item5	0.804			
DS		0.867	0.909	0.715
Item1	0.813			
Item2	0.820			
Item3	0.895			
Item4	0.852			
DIP		0.916	0.932	0.630
Item1	0.779			
Item2	0.806			
Item3	0.780			
Item4	0.812			
Item5	0.759			
Item6	0.782			
Item7	0.828			
Item8	0.803			
DTO		0.892	0.920	0.698
Item1	0.867			
Item2	0.852			
Item3	0.803			
Item5	0.830			
Item6	0.823			

Table 2. Construct reliability and Validity. $N = 348$; Ca, Cronbach's alpha; CR, composite reliability; AVE, average variance; DTL, digital transformational leadership; DS, digital strategy; DIP, digital innovation performance; DTO, digital transformation openness.

Models	Factor	χ^2	df	χ^2/df	RMSEA	CFI	TLI	SRMR
Four-factor model	DTL, DS, DIP, DTO	317.802	203	1.566	0.040	0.974	0.970	0.043
Three-factor model	DTL + DS, DIP, DTO	801.334	206	3.890	0.091	0.863	0.847	0.072
Two-factor model	DTL + DS + DIP, DTO	1322.782	208	6.360	0.124	0.744	0.716	0.103
Single-factor model	DTL + DS + DIP + DTO	1963.626	209	9.395	0.155	0.598	0.555	0.122

Table 3. Results of confirmatory factor analysis. Note: $N = 348$; DTL, digital transformational leadership; DS, digital strategy; DIP, digital innovation performance; DTO, digital transformation openness.

For discriminant validity, compared to other competition models, the theoretical four-factor model (digital transformational leadership, digital strategy, digital innovation performance, and digital transformation openness) had a better fit to the data [$\chi^2/df = 1.566$, (CFI) = 0.974, (TLI) = 0.970, (RMSEA) = 0.040, and (SRMR) = 0.043 (see Table 3)]. The CFA results showed that the theoretical four-factor model had satisfactory discriminant validity.

Furthermore, the heterotrait–monotrait ratio of correlations (HTMT) criteria were employed to test the discriminant validity. Different recommendations exist for confirming the HTMT criterion, with the conservative criterion suggesting that the HTMT value should be below 0.85⁵⁷, and the classical criterion indicating that the HTMT value should be below 0.90⁵⁸. The HTMT ratio table demonstrates that all values fall within the range of 0.422 to 0.595, which is lower than the specified criterion, thus confirming discriminant validity Table 4i.

Common method variance

Common method variance (CMV) may affect the empirical results because our study data were collected through self-report questionnaires. Podsakoff et al.⁵⁹ showed that procedural and statistical techniques can be adopted for CMV. In the statistical technique, the possibility of common method bias was tested using Harman's one factor test⁶⁰. A principal component factor analysis with varimax rotation was used on the items of digital transformational leadership, digital strategy, digital innovation performance, and digital transformation

	DIP	DTL	DS	DTO
DIP				
DTL	0.461			
DS	0.595	0.503		
DTO	0.422	0.464	0.522	

Table 4. Heterotrait–monotrait ratio ($N = 348$).

Variables	Mean	SD	1	2	3	4	5	6	7	8	9
1. Gender	1.50	0.501	1								
2. Age	2.60	1.100	−0.049	1							
3. Education	3.07	0.897	0.049	−0.093	1						
4. Seniority	2.67	1.128	−0.094	0.475**	0.021	1					
5. Years	2.92	1.266	−0.007	−0.165**	−0.073	0.075	1				
6. DTL	3.77	0.751	0.056	0.042	−0.055	0.034	0.002	1			
7. DS	3.93	0.838	0.073	0.055	−0.029	0.014	0.025	0.436**	1		
8. DIP	3.91	0.758	0.034	0.142**	0.006	0.089	−0.088	0.412**	0.530**	1	
9. DTO	3.77	0.852	0.067	−0.024	−0.080	−0.035	−0.023	0.408**	0.459**	0.382**	1

Table 5. Means, standard deviations (SD), and correlations. Note: $N = 348$; ** $p < 0.01$; * $p < 0.05$; DTL, digital transformational leadership; DS, digital strategy; DIP, digital innovation performance; DTO, digital transformation openness.

openness. This result revealed multiple factors with eigenvalues greater than 1. The first factor accounted for 22.96% ($< 50\%$) loading, which proved the absence of CMV⁶¹.

Further, we conducted the unmeasured latent method factor⁵⁹, to test CMV. A comparison of the latent method factor model ($\chi^2/df = 1.276$, CFI = 0.977, TLI = 0.971, RMSEA = 0.040, SRMR = 0.043) and the four-factor model ($\chi^2/df = 1.566$, CFI = 0.974, TLI = 0.970, RMSEA = 0.040, SRMR = 0.043) indicated no significant changes in CFI⁶². Thus, CMV was not a major problem for the data.

Means and correlations

The descriptive statistics and correlation analysis results presented in Table 5 indicate that digital transformational leadership is positively correlated to digital strategy ($r = 0.436$, $p < 0.05$), digital innovation performance ($r = 0.412$, $p < 0.05$), and digital transformation openness ($r = 0.408$, $p < 0.05$). Digital strategy and digital transformation openness are positively correlated to digital innovation performance ($r = 0.530$, $p < 0.05$ and $r = 0.382$, $p < 0.05$). The correlation between the key variables supports our hypotheses on the direct and indirect effects of digital transformational leadership on digital innovation performance.

Structural model

Before testing the structural model, we first tested its collinearity. Collinearity is measured using the variance inflation factor (VIF), and, ideally, the VIF values should be close to and lower than 3⁵³. The results show that all VIF values are below this threshold, suggesting no collinearity among the constructs. We also examined the R^2 value, which indicates the model's predictive power by showing the endogenous variable's variance that the exogenous variables can explain. The R^2 value for DIP (0.415) indicates that all the constructs combined explain 41.5% of the variance in DIP. The R^2 value for DS is 0.199. Further, we checked the Q^2 values to assess the predictive relevance values generated by the variables. The Q^2 values for DIP (0.260) and DS (0.191) were above 0, which means that the model has predictive relevance.

To examine the hypotheses, bootstrapping was carried out using SmartPLS 4.1 with 10,000 subsamples based upon percentile bootstrapping with a two-tailed test type and a significance level of 0.05. The PLS-SEM bootstrapping approach statistically determined the structural model coefficients representing the hypothesized relationships. Simultaneously, to ensure the robustness of the results, we included age, gender, education, seniority, and years implementing DT as control variables in the structural model assessment.

Direct effect and mediation effect testing

Figure 2; Table 6 portray the results of the structural path analysis. The results show that digital transformational leadership has a significant positive impact on digital innovation performance ($B = 0.146$, $p < 0.05$; 95% CI: 0.039–0.263; $T = 2.535$), supporting Hypothesis 1. And, digital strategy has a significant positive impact on digital innovation performance ($B = 0.454$, $p < 0.001$, 95% CI: 0.339–0.554; $T = 8.296$), supporting Hypothesis 2.

Further, the results show that digital strategy significantly mediates the relationship between digital transformational leadership and digital innovation performance ($B = 0.203$, $p < 0.001$, 95% CI: 0.136–0.274; $T = 5.734$). Therefore, Hypotheses 3 is supported.

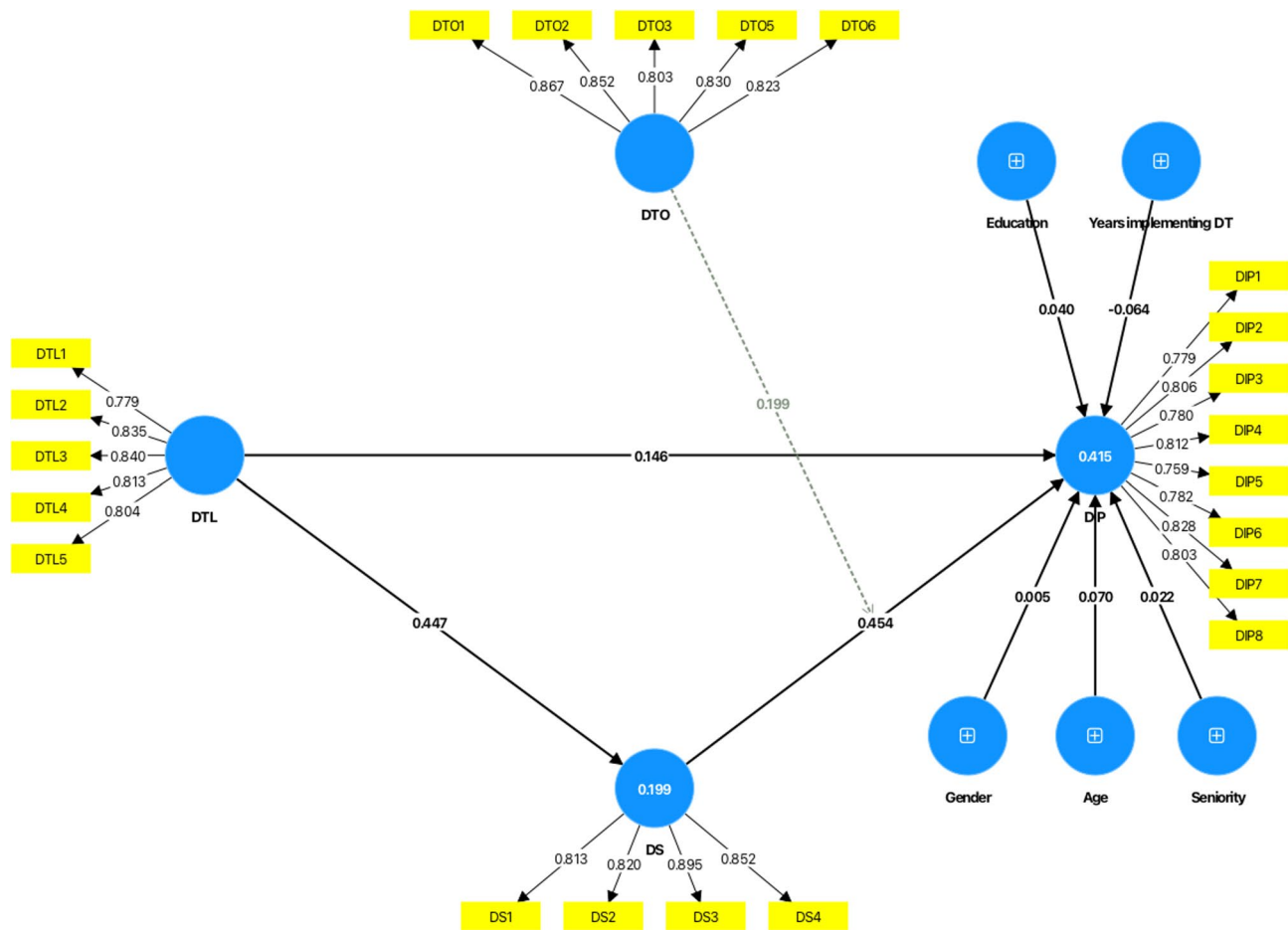


Fig. 2. PLS path model from SmartPLS.

Hypotheses	Relationship	B	Std Dev	T Statistics	P-Value	LLCI	ULCI	Results
Direct effect								
	DTL -> DS	0.447	0.052	8.649	0.000	0.343	0.544	Supported
H1	DTL -> DIP	0.146	0.057	2.535	0.011	0.039	0.263	Supported
H2	DS -> DIP	0.454	0.055	8.296	0.000	0.339	0.554	Supported
Mediation effect								
H3	DTL -> DS -> DIP	0.203	0.035	5.734	0.000	0.136	0.274	Supported
Moderation effect								
H4	DTO x DS -> DIP	0.199	0.055	3.594	0.000	0.079	0.298	Supported

Table 6. Hypothesis testing.

Moderating effect testing

We examined the moderating effect of digital transformation openness on the relationship between digital strategy and digital innovation performance. As shown in Fig. 2; Table 6, the interaction between digital strategy and digital transformation openness is significantly and positively related to digital innovation performance ($B = 0.203$; $p < 0.001$; 95% CI: 0.079–0.298; $T = 3.594$), indicating that digital transformation openness positively moderates the relationship between digital strategy and digital innovation performance. Hence, Hypothesis 4 is supported. According to the suggestions of (Toothaker, 1994), this study further analyzed the moderating effect by testing the simple slopes at high and low levels of digital transformation openness, and the moderating effect diagram was drawn (see Fig. 3).

Moderated mediation testing

The index of moderated mediation⁶³ presented in Table 7 indicates the total moderated mediation effect. The effect was significant ($B = 0.105$; $p < 0.01$; 95% CI: 0.053–0.162; $T = 3.131$), indicating that the indirect effect

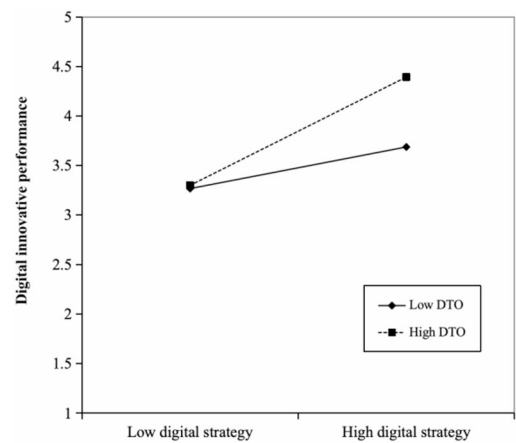


Fig. 3. Moderating effect of Digital Transformation Openness on the relationship between Digital Strategy and Digital Innovation Performance.

	DTO	B	Std Dev	T Statistics	P-Value	LLCI	ULCI
DTL -> DS -> DIP	High(+ 1SD)	0.288	0.056	5.108	0.000	0.182	0.404
	Middle	0.199	0.038	5.242	0.000	0.128	0.275
	Low(-1SD)	0.109	0.036	3.001	0.003	0.041	0.184
Index of conditional mediation		0.105	0.033	3.131	0.001	0.053	0.162

Table 7. Results of moderated mediation effect.

of digital transformational leadership on DIP through digital strategy was moderated by digital transformation openness.

The conditional indirect effect on the values of the moderator was calculated, namely the mean, one standard deviation above, and one standard deviation below. The results are shown in Table 7. The model of digital transformational leadership influencing digital innovation performance via digital strategy shows that at low levels of digital transformation openness, the mediating effect is significant ($B=0.109$; $p<0.01$; 95% CI: 0.036–0.183; $T=3.001$). At higher levels of openness, the mediating effect becomes even stronger ($B=0.288$; $p<0.001$; 95% CI: 0.182–0.404; $T=5.108$). The analysis results show that the higher the digital transformation openness, the stronger the mediating role of digital strategy in the relationship between digital transformational leadership and digital innovation performance, thus supporting Hypothesis 5.

Discussion
Discussion of findings

Grounded in RBV, this study develops a theoretical model incorporating mediating and moderating pathways to systematically investigate how digital transformational leadership affects firms’ digital innovation performance through digital strategy, while also examining the moderating role of digital transformation openness. By conducting in-depth empirical analysis, the study validates the key hypothesized paths in the model, uncovering complex and dynamic mechanisms linking digital leadership, strategy, and performance.

First, the study confirms that digital transformational leadership significantly enhances firms’ digital innovation performance. This finding aligns with existing literature highlighting the pivotal role of digital transformational leadership in driving organizational change. Digital transformational leaders demonstrate strong digital awareness and strategic foresight, enabling them to identify emerging technological trends, coordinate resources, and stimulate employees’ digital thinking and creativity, thereby accelerating innovation speed and improving outcomes within the digital environment. From the RBV, digital transformational leadership is viewed as a rare and valuable form of human capital that allows organizations to coordinate internal capabilities and produce innovation outcomes, ultimately supporting long-term competitive advantage.

Second, the study reveals that digital strategy partially mediates the relationship between digital transformational leadership and digital innovation performance. This suggests that the impact of leadership on innovation outcomes is indirect and contingent upon the coordinated implementation of strategic systems and the restructuring of organizational processes. This finding aligns with Bharadwaj et al.¹³ three-layer digital strategy framework, which posits that digital technology must be embedded within business processes, organizational structure, and the value chain to translate into tangible performance outcomes. Furthermore, the study highlights that through leaders’ empowerment mechanisms, digital strategy not only conveys the organization’s future vision but also offers guidance and institutional support for employees’ innovative activities.

This mediation mechanism reflects the RBV that strategic capabilities, such as digital strategy, allow firms to reconfigure and integrate resources, thereby transforming digital potential into actual performance.

Third, digital transformation openness significantly moderates the relationship between digital strategy and digital innovation performance. This implies that employees' open and supportive attitudes toward change enhance the execution efficiency and outcome conversion of digital strategy. This mechanism reveals profound interactions among "leader, strategy, and performance," echoing Subramaniam et al.⁶⁴ assertion on the significant influence of "digital cognitive engagement" on innovation outcomes. The study further finds that digital transformation openness amplifies the indirect effect of digital transformational leadership on performance via the strategic pathway. Specifically, employees' attitudes not only influence strategy implementation but also magnify the indirect effectiveness of leadership. This finding underscores the crucial role of employees' proactive agency in organizational change, corroborating Vial⁶⁵ claim that cultural elements are indispensable to digital transformation.

Fourth, although the moderated mediation effect was statistically significant, the effect size was relatively small. This suggests that although openness to digital transformation can enhance the positive effect of digital strategy on employees' innovative performance, the facilitating role of openness becomes more evident only when it surpasses a certain threshold.

Theoretical implications

First, drawing on the Resource Based View (RBV) theory⁷, this study reveals the internal mechanisms through which firms integrate and utilize digital resource portfolios to gain competitive advantage. Furthermore, the study finds that digital transformation leadership, as a unique managerial resource, is positively associated with digital innovation performance, and that digital strategy functions as a critical channel for resource orchestration within this relationship. Accordingly, this study contributes to the RBV literature by deepening our understanding of how human resources drive value creation in the context of digital transformation.

Second, despite extensive research on traditional transformational leadership, systematic investigations of Digital Transformational Leadership in the context of digital transformation remain limited. This study integrates theories of traditional and digital leadership to define and empirically examine the pathways through which Digital Transformational Leadership influences digital innovation performance. It emphasizes DTL's vital role in helping organizations navigate environmental uncertainty, foster technology adoption, and reshape organizational culture⁹. The study broadens the scope of leadership theory by responding to Benitez et al. (2022)'s call for a redefinition of leadership in the digital era, while clarifying the leader's role in legitimizing organizational change⁶⁶.

Third, although previous research has confirmed the positive effects of leadership behaviors on organizational performance, the underlying mechanisms driving this transformation remain insufficiently explored. This study employs digital strategy as a mediating variable, focusing on how leadership enhances performance via strategic deployment and restructuring of business processes. The study confirms Bharadwaj et al.¹³ assertion that the strategic layer constitutes a critical channel for technology-enabled performance, aligning with Alnuaimi et al.⁹ claim that strategic alignment is essential for effective digital leadership.

Fourth, although recent years have seen growing recognition of employees' subjective role in organizational change, empirical evidence on how "openness" moderates the strategy–performance relationship remains limited. This study introduces "digital transformation openness" as a moderating variable and empirically confirms its significant impact on the effectiveness of digital strategy, representing a critical manifestation of employees' digital cognition and behavioral engagement⁶⁴. In high-openness environments, digital strategies are implemented more effectively, and employee collaboration is strengthened, thereby improving organizational performance. This finding also addresses the theoretical gap identified by Vial⁶⁵ regarding the mechanism through which digital culture shapes transformation outcomes.

Practical implications

First, the findings indicate that managers, acting as organizational agents, can influence key factors that improve digital innovation performance, thereby benefiting both the organization and its employees. Management has significant authority over the development and implementation of digital transformational leadership capabilities. Leadership development programs that emphasize digital transformation skills such as data-driven decision making, digital ecosystem thinking, and technology-based empowerment can better prepare managers to foster innovative digital environments. These leaders can effectively navigate their teams through the complexities of digital transformation and convert technological potential into tangible innovation outcomes.

Second, the design of digital strategy systems falls within the domain of management influence. Our findings highlight that digital strategy acts as a key mediating mechanism through which leadership affects innovation performance. Managers should develop comprehensive digital strategies that extend beyond technology adoption to include systematic business model transformation, process digitization, and value chain restructuring. An effective approach to implementing digital strategy is to initially focus on organizational processes that best align with existing digital capabilities. The resulting sense of competence, clarity, and technological mastery facilitates the transformation. For example, organizations with strong data analytics capabilities might prioritize digital strategy elements that utilize data-driven insights to drive product innovation.

Third, organizations should place greater emphasis on fostering an innovative culture. The organizational culture for digital transformation openness may take longer than structural elements (such as digital infrastructure or formal digital policies) to change. In environments with limited digital openness, employees are more likely to preserve existing routines, as innovation often entails elevated perceived risks and psychological stress. Even under strong leadership, strategic initiatives may fail to produce tangible innovation outcomes. In contrast, cultural transformation efforts are more likely to deliver meaningful outcomes only when digital

openness exceeds a moderate level, such as when digital engagement and adaptive practices are deeply embedded in everyday workflows. Practitioners should evaluate the baseline level of organizational openness and tailor interventions accordingly, emphasizing capability development, psychological safety, and digital communication channels to cultivate a culture of adaptability and openness.

Limitations and future research

First, this study used a time lag data collection approach to reduce common method variance (CMV) and better examine causal relationships between variables across time. However, since all variables were reported by employees themselves, concerns regarding CMV may still remain. Future studies may consider using data collected from multiple time points and multiple sources. In particular, when assessing digital innovation performance, using objective indicators from company records would be helpful in mitigating CMV more effectively.

Second, the sample for this study was exclusively drawn from manufacturing firms located in Shandong Province, China. Although these firms vary in size and digital development maturity, the sample is still limited in terms of industrial scope and cultural context. Future research should further examine the generalizability of the findings across countries, regions, and different industry sectors. A cross-national and cross-industry comparative perspective may help explore how digital leadership influences strategic implementation and innovation performance under varying conditions, thereby enhancing the external validity and universality of the conclusions. Moreover, future studies should examine whether the effectiveness of digital transformational leadership differs across specific digital technologies, as technologies such as artificial intelligence, IoT, and blockchain may require distinct leadership responses and strategic configurations.

Third, although this study introduced digital transformation openness as a moderating variable and obtained initial findings, several influential contextual factors remain unaccounted for in the model. These include organizational learning capacity, corporate culture, and the maturity of technological infrastructure, all of which could significantly shape the functioning of leadership in digital contexts. Future research may expand the scope of moderating or mediating variables and construct more complex interaction models to deepen insights into the collaborative dynamics among individuals, strategy, and technology during digital transformation.

Fourth, although this study included several individual-level control variables, such as age, gender, education, seniority, and years of digital transformation implementation, it did not incorporate potentially relevant organizational-level controls. Organizational characteristics such as firm size, R&D intensity, and market competition may also affect the adoption of digital strategies and innovation performance. However, constraints related to data availability and concerns about model over-control prevented their inclusion in this study. Future research should integrate multilevel control variables and employ hierarchical or cross-level modeling techniques to better rule out alternative explanations and enhance the robustness of findings within organizationally nested data structures.

Data availability

Data are available upon reasonable request from the corresponding author.

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Author contributions

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Declarations

Ethics Statement (Methods)

As protection of all participants, all subjects provided informed consent before participating in this study and voluntarily made their decision to complete surveys. The protocol was approved by an institutional review board in Kangnam University of Korea. All methods were carried out in accordance with relevant guidelines and regulations.

Competing interests

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Additional information

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